

Supplementary Material

PanoDive360: A Novel Equirectangular Projection Dataset and Enhanced Attention Mechanism for Underwater 360-Degree Multiclass Semantic Segmentation

Ghulam Arbi, Lu Zhang, and Yanyong Zhang
University of Science and Technology of China
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This supplement records the reproducibility protocol and the extra experiments that do not fit in the 6-page ACM main paper: random-seed robustness (seeds 42, 123, 777), training/validation curves, and ERP video-inference cost. All numbers and graphs match the PanoDive360 chapter. Upload this folder to <https://github.com/aiguo112/PanoDive360> as supplementary/. For CMT double-blind review, strip the author block before uploading the PDF.

1. Reproducibility protocol

1.1 Software and hardware

All models were implemented in Python 3.8.19 with PyTorch 1.8.0, CUDA 11.1 and cuDNN 8902. Training used dual NVIDIA GeForce RTX 3060 GPUs (12 GB each) with data parallelism. Latency and FPS were measured on an RTX 5070 Ti: batch size = 1, 20 warm-up iterations, 100 timed iterations, CUDA synchronised, input padded to 1440×736.

1.2 Training hyperparameters

The protocol below was applied identically to U-Net, U-Net++, PSPNet, DeepLabv3+ and ERP-CBAM. The paper resolution is 1440×720 (2:1 ERP, no crop). The public `panodive360/config.py` currently defaults to a square 1024×1024 resize; that setting does not reproduce the paper tables. Override it to (1440, 720) before rerunning.

Hyperparameter	Value
Classes	11
Batch size	4
Optimizer	Adam ($\beta_1=0.9$, $\beta_2=0.999$)
Learning rate	1×10^{-4}
Loss	CE + Dice
Epochs / early-stop patience	25 / 5
LR scheduler	ReduceLROnPlateau (factor 0.5, patience 3)
Input (train/test)	1440×720 (2:1 ERP)
Video inference (padded)	1440×736
Train / val / test	1,472 / 421 / 211
Backbone	ResNet-34 (ImageNet)

Table S1. Training hyperparameters on PanoDive360.

1.3 Random seeds

The nine-metric comparison against the four baselines uses a single run so that the ranking is exactly reproducible (main seed 42). ERP-CBAM was then retrained twice more under identical hyper-parameters, data splits, and the 25-epoch protocol, using seeds 123 and 777. Python, NumPy and PyTorch (CPU and CUDA) generators were seeded consistently; cuDNN was set deterministic. The epoch-25 checkpoint was evaluated on the same 211-image test set. These runs share one train/val/test split: they probe seed robustness, not split variance.

Helper in scripts/seed.py:

```
set_seed(42) # or 123, 777
```

```
Then: python train_seeded.py --seed 42 (repeat for 123 and 777).
```

2. Three-seed robustness

Table S2 reports test mIoU and pixel accuracy for the three seeds. The seed-42 mIoU of 0.6702 lies within ordinary run-to-run variation of the main-table value 0.662. The lowest seed (123, mIoU 0.6342) still exceeds DeepLabv3+ (0.536) by 18.3% relative.

Seed	Test mIoU	Pixel accuracy
42	0.6702	0.9597
123	0.6342	0.9592
777	0.6465	0.9578
Mean $\pm$ Std	0.6503 $\pm$ 0.0183	0.9589 $\pm$ 0.0010

Table S2. ERP-CBAM (DeepLabv3+, ResNet-34) on the same 211-image test split.

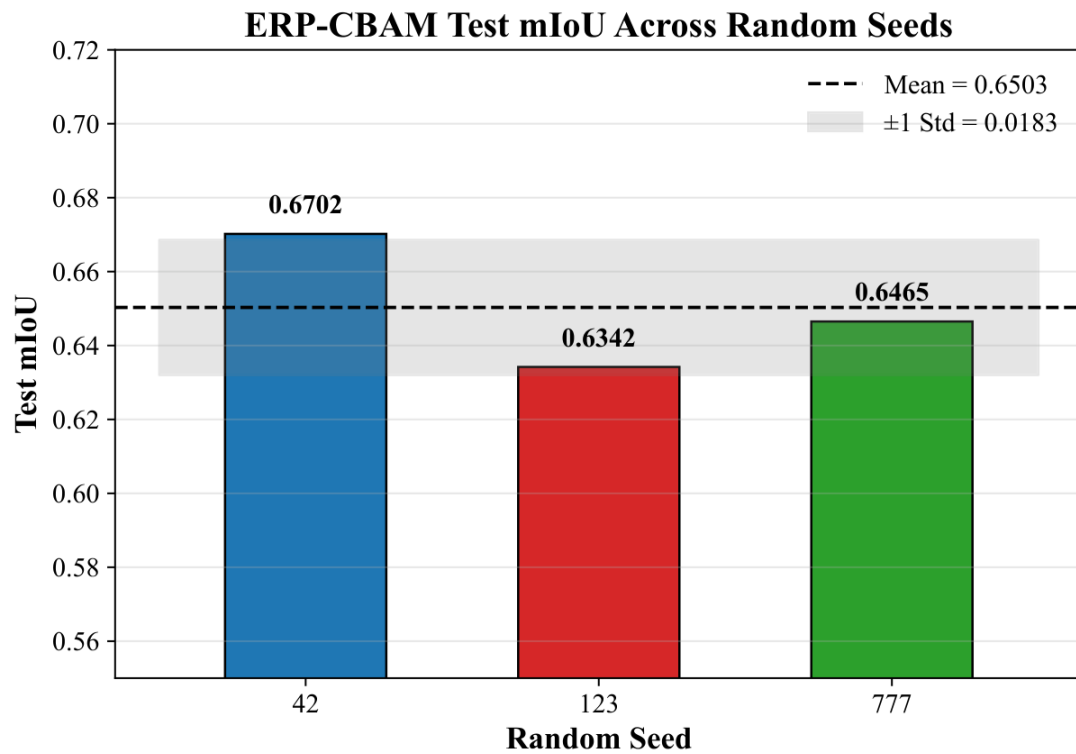


Figure S1. Test mIoU of ERP-CBAM for seeds 42, 123 and 777. Dashed line: mean (0.6503). Band: ± 1 standard deviation (0.0183).

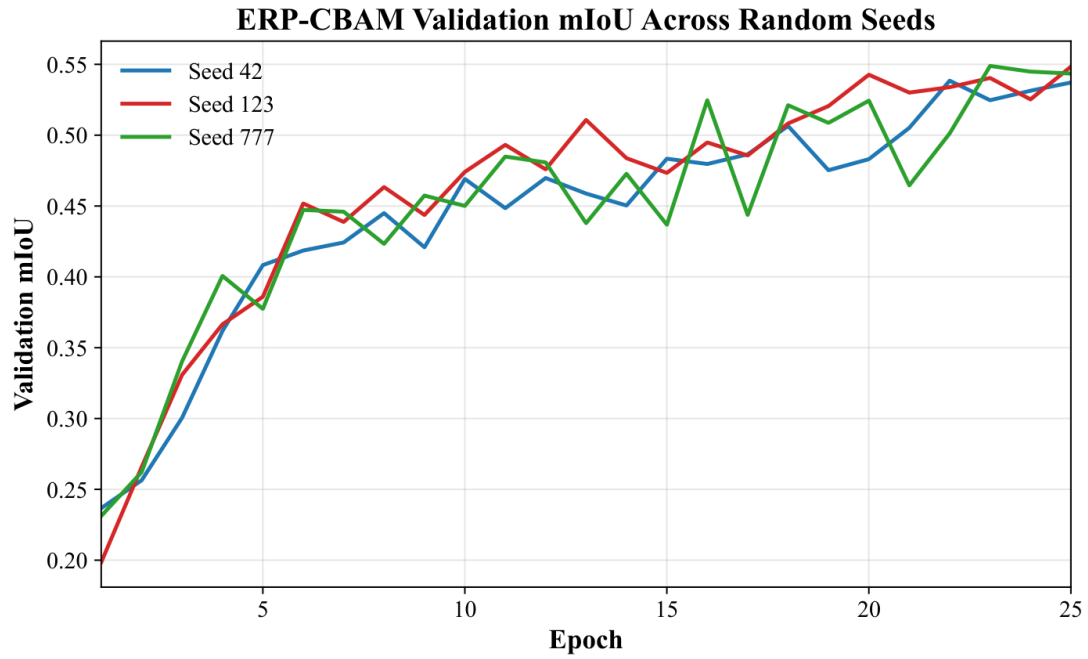


Figure S2. Validation mIoU over 25 training epochs for the three seeded ERP-CBAM runs. The trajectories remain closely aligned.

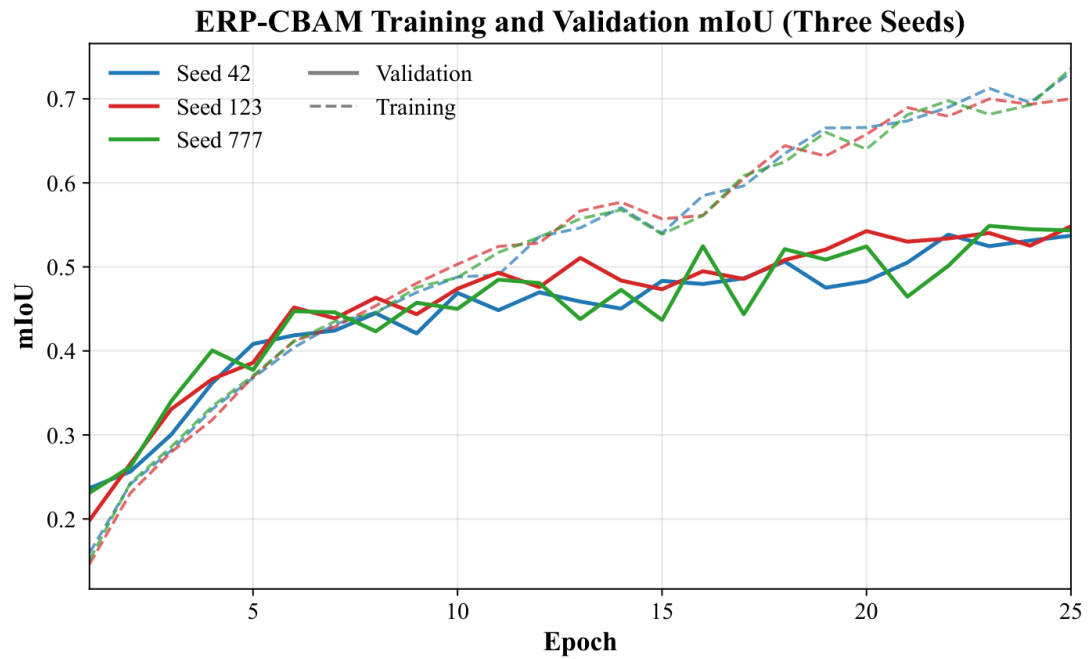


Figure S3. Training and validation mIoU for the three seeds (solid: validation; dashed: training). Extra curve not shown in the main paper.

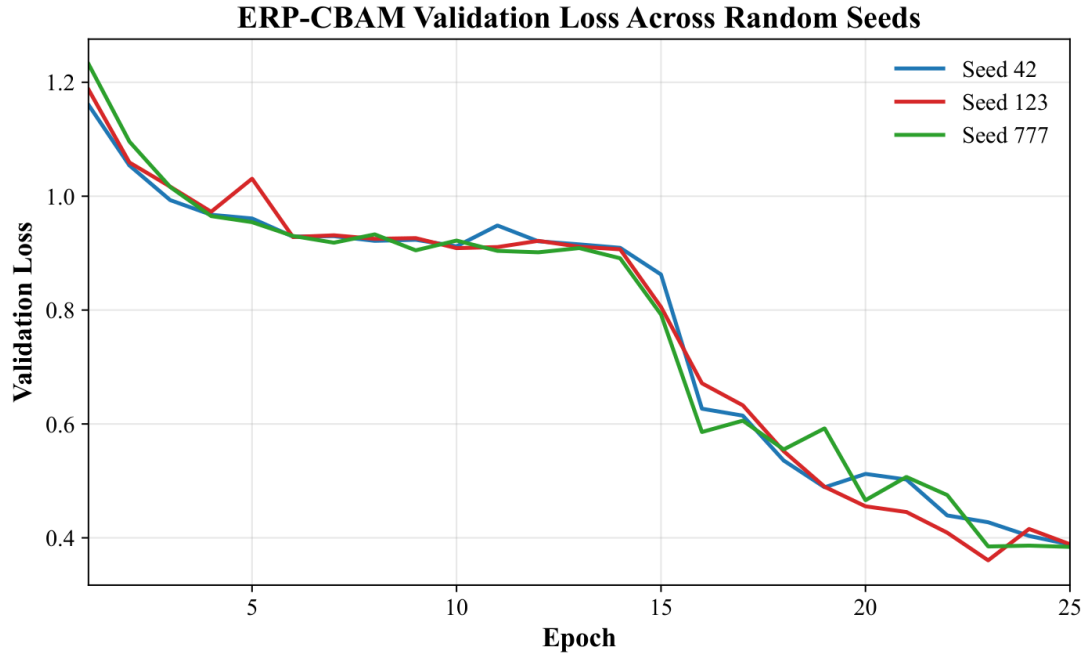


Figure S4. Validation loss over 25 epochs for seeds 42, 123 and 777. Extra curve not shown in the main paper.

3. Computational cost of ERP video inference

ERP-CBAM is DeepLabv3+ with latitude-weighted spatial attention. It cannot be substantially faster than its host. Table S3 shows +0.012 M parameters (+0.05%) and +0.017 GFLOPs (+0.007%) at 1440×736 . On the RTX 5070 Ti, ERP-CBAM measures 79.1 FPS (12.64 ms) against 86.8 FPS for DeepLabv3+ (11.53 ms), about 9% slower. The claim that stands is that the mIoU gain is obtained inside the DeepLabv3+ compute envelope, not that ERP-CBAM is the fastest model.

Model	Params (M)	GFLOPs	Peak mem. (MB)	Latency (ms)	FPS
U-Net	24.44	255.8	623	16.57	60.3
U-Net++	26.08	597.6	1536	38.46	26.0
PSPNet	21.48	77.2	460	4.83	206.9
DeepLabv3+	22.44	255.3	774	11.53	86.8
ERP-CBAM	22.45	255.4	861	12.64	79.1

Table S3. ERP video inference (1440×736 , batch = 1, 11 classes). Params/GFLOPs are hardware-independent. Latency/FPS: RTX 5070 Ti.

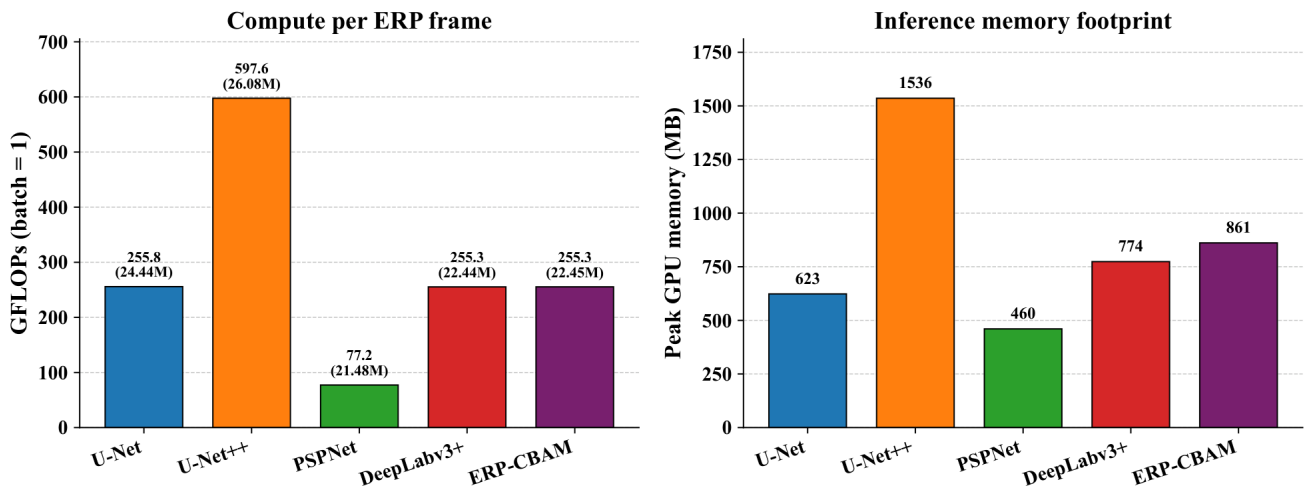


Figure S5. Left: GFLOPs per ERP frame (parameter count in parentheses). Right: peak GPU memory. ERP-CBAM is indistinguishable from DeepLabv3+ on FLOPs and stays below 1 GB of inference memory.

4. Full metric comparison

Table S4 repeats the nine-metric test-set comparison. Mean IoU and Balanced Accuracy are the primary ranking metrics because PanoDive360 is strongly imbalanced (diver in 1,010 of 1,052 frames; dolphin in 27). ERP-CBAM is not best on accuracy, precision, or MCC; treat it as a diagnostic ERP baseline rather than a generally superior architecture.

Model	Acc.	mIoU	Prec.	Rec.	F1	Dice	Spec.	Bal.Acc.	MCC
U-Net	0.961	0.444	0.741	0.672	0.663	0.663	0.983	0.828	0.680
U-Net++	0.959	0.437	0.768	0.647	0.646	0.646	0.981	0.814	0.674
PSPNet	0.923	0.371	0.780	0.470	0.461	0.461	0.966	0.718	0.519
DeepLabv3+	0.947	0.536	0.780	0.680	0.655	0.655	0.979	0.830	0.726
ERP-CBAM	0.955	0.662	0.773	0.758	0.726	0.831	0.983	0.870	0.716

Table S4. PanoDive360 test set (single run, seed 42).

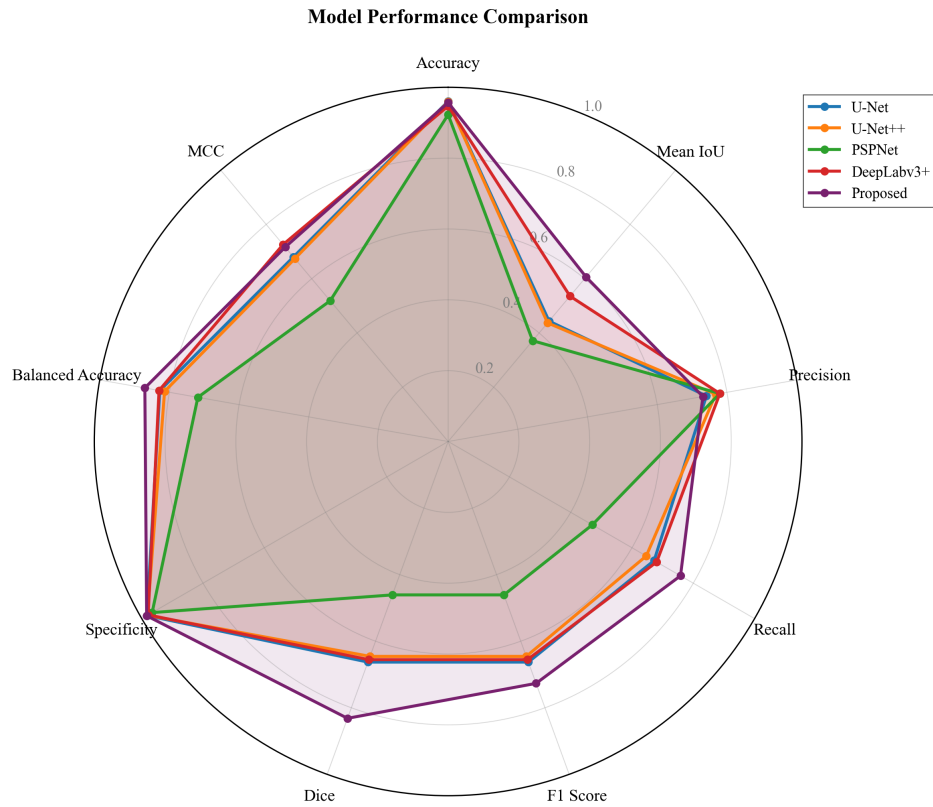


Figure S6. Radar comparison of the five CNNs on the PanoDive360 test set.

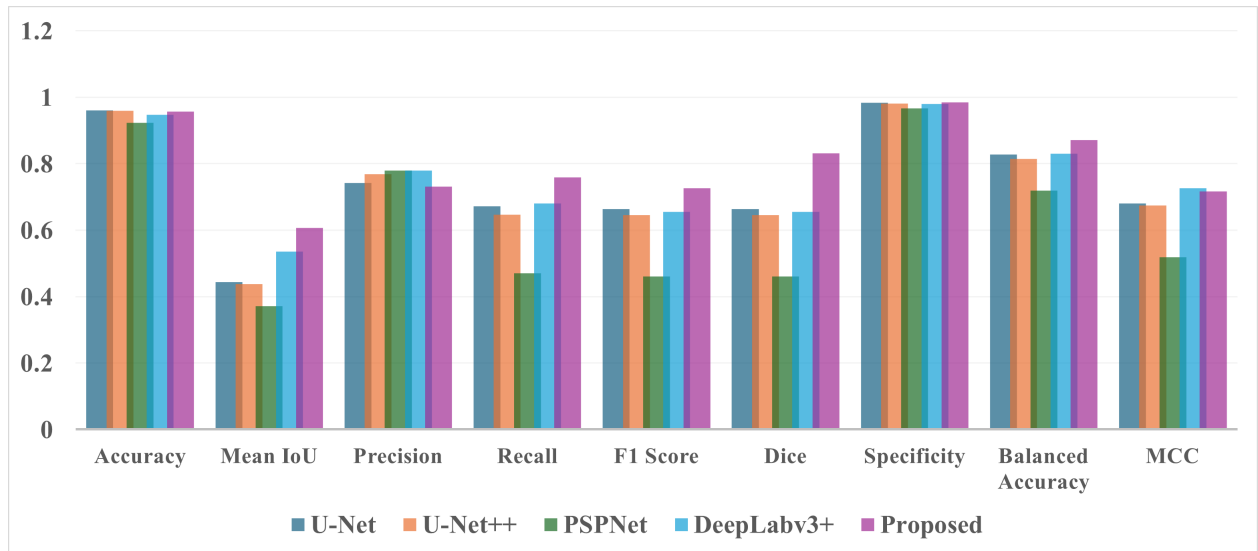


Figure S7. Grouped bar chart of the same nine metrics.

5. Per-class IoU

Dolphin (IoU 0.107) is a data-limited outlier (27 instances), not an architectural failure. Sea turtle remains strong (IoU 0.845) despite only 41 instances.

Class	IoU	Precision	Instances
Background	0.958	0.982	1,025
Diver	0.741	0.856	1,010
Shark	0.838	0.909	310
Fish	0.499	0.625	267
Sea Turtle	0.845	0.875	41
Dolphin	0.107	0.335	27
Sea Lion	0.779	0.901	68
Coral	0.561	0.787	96
Shipwreck	0.790	0.879	275
Seaweed	0.606	0.703	323
Rock	0.564	0.662	272

Table S5. Per-class IoU and precision for ERP-CBAM. Instance counts are dataset-wide.

6. Ablation

Standard CBAM without latitude weighting decreases mIoU from 0.536 to 0.409. Latitude weighting is the main driver of the balanced-metric gains (recall 0.723 $\rightarrow$ 0.758, balanced accuracy 0.851 $\rightarrow$ 0.870).

Variant	Acc.	mIoU	Rec.	F1	Dice	Bal.Acc.	MCC
DeepLabv3+	0.947	0.536	0.680	0.655	0.655	0.830	0.726
Baseline + CBAM	0.920	0.409	0.544	0.543	0.824	0.752	0.525
ERP-CBAM w/o ERP	0.956	0.635	0.705	0.751	0.907	0.839	0.737
ERP-CBAM w/o Lat.	0.959	0.637	0.723	0.750	0.908	0.851	0.737
ERP-CBAM w/o CA	0.951	0.569	0.652	0.690	0.896	0.815	0.690
ERP-CBAM (full)	0.955	0.662	0.758	0.726	0.831	0.870	0.716

Table S6. ERP-CBAM ablation on the PanoDive360 test set.

7. Dataset class inventory

PanoDive360 contains 1,052 pixel-annotated monoscopic ERP stills from 30 publicly shared 4K 360-degree videos, 11 classes, 2:1 aspect. The full image archive is not in git; six labelled examples are in data/examples/. RGB colours match panodive360/config.py.

Class	RGB	Instances	% of 1,052
Background	(0, 0, 0)	1,025	97.4
Diver	(167, 242, 82)	1,010	96.0
Shark	(166, 3, 3)	310	29.5
Fish	(255, 237, 29)	267	25.4
Sea Turtle	(255, 0, 243)	41	3.9
Dolphin	(30, 95, 170)	27	2.6
Sea Lion	(169, 205, 248)	68	6.5
Coral	(106, 37, 163)	96	9.1
Shipwreck	(115, 76, 20)	275	26.1
Seaweed	(233, 180, 245)	323	30.7
Rock	(245, 94, 94)	272	25.9

Table S7. Class colours and instance counts.

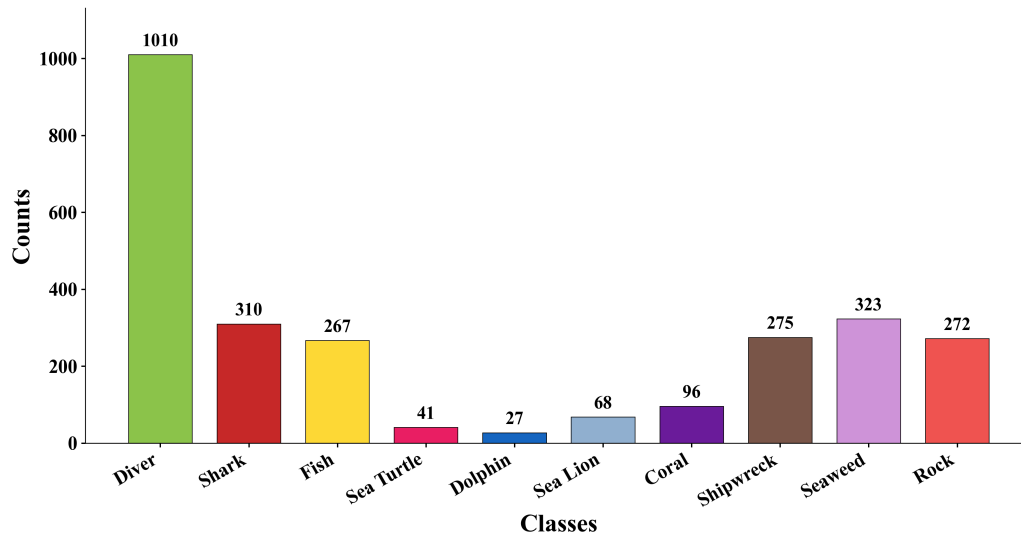


Figure S8. Class instance distribution. Diver dominates (1,010); dolphin (27) and sea turtle (41) are rare.

8. What to upload

Copy this entire folder to the PanoDive360 repository as `supplementary/`. Optionally paste the short section in `README_GITHUB_SNIPPET.md` into the root `README`. If MMAAsia review is still double-blind, do not put the GitHub URL in the submitted PDF; upload only `PanoDive360_supplementary.pdf` to CMT with the author block removed.